TEXORA · LECTURE SHEET

Trims and Accessories Calculation

Sewing thread by stitch-type consumption ratios, buttons by the gross, labels, polybags and cartons: calculating every non-fabric item of the 12,000-shirt order so that production never stops for a missing trim.

AM-3207 · ADVANCED MACHINERIES AND RELEVANT CALCULATIONS · TOPIC 4 · SPECIALIZATION — APPAREL MANUFACTURING

1. Why trims are calculated, not estimated

Fabric is the largest cost in a garment, but trims cause a disproportionate share of stoppages and claims: a line halts for want of a two-cent button, and a shipment is rejected for a wrong care label. Every trim is therefore calculated from the garment's bill of materials — quantity per garment × order quantity × a wastage allowance — and bought in the supplier's trading unit (cones, gross, cartons). The calculations below use the course order: 12,000 woven shirts.

2. Sewing-thread calculation

Thread consumption is calculated seam by seam. For each seam: thread used = seam length × the **consumption ratio** of its stitch type — the metres of thread locked into one metre of seam, including needle and looper/bobbin thread and the stitch geometry.

STITCH TYPE	USE ON THE SHIRT	THREAD RATIO (PER UNIT SEAM LENGTH)
301 lockstitch	Main construction seams, topstitching	2.5
401 chainstitch	Side seams on some constructions, feed-off-arm seams	4.0
504 overlock (3-thread)	Edge neatening of cut parts	12.0

Ratios are standard planning values for typical stitch lengths; a factory verifies them by the unravelling test — marking a measured seam length, unpicking it and measuring the thread recovered.

WORKED EXAMPLE — THREAD FOR 12,000 SHIRTS

Seam map per shirt: lockstitch seams total **260 cm**; overlock seams total **60 cm**.
Lockstitch thread = 260 × 2.5 = 650 cm. Overlock thread = 60 × 12 = 720 cm. Total per garment = 1,370 cm = **13.7 m**.
With 10% sewing and cone-end wastage: 13.7 × 1.10 = **15.07 m per garment**.
Order total = 15.07 × 12,000 = **180,840 m**. In 5,000 m cones: 180,840 ÷ 5,000 = 36.2 → **buy 37 cones** of the main shade (each additional shade is calculated the same way from its own seam share).



Figure 1 — The overlock seam is shorter on the garment but consumes more thread than all lockstitch seams together: stitch geometry, not seam length, drives thread cost.

3. Buttons, labels and packing materials

WORKED EXAMPLE — BUTTONS, LABELS, POLYBAGS, CARTONS

Buttons. The shirt sews 9 buttons (6 front, 1 collar stand, 2 cuff) plus 1 spare in the polybag = 10 per garment. Order: 10 × 12,000 = 120,000; with 3% loss and breakage: 120,000 × 1.03 = **123,600 buttons**. Buttons trade by the gross (144 pieces): 123,600 ÷ 144 = 858.3 → **buy 859 gross**.
Labels. Main label, size label and care label — 3 per garment, each calculated separately: 12,000 × 1.02 = **12,240 of each type**.
Polybags. 1 per garment × 1.01 = **12,120 bags**.
Cartons. 24 shirts per export carton: 12,000 ÷ 24 = **500 cartons** (plus a small spare quantity for repacking, ordered to the packing list).

4. Wastage allowances — why each trim differs

Allowance percentages are not arbitrary: thread carries the highest (10%) because cones end mid-seam, seams are unpicked and re-sewn, and shade lots must not be mixed; buttons carry breakage and sewing-machine loss (3%); labels carry cutting and miscount loss (2%); polybags carry the least (1%). Applying one blanket percentage to all trims either starves the line (too little thread) or strands money in leftover stock (too many buttons). The allowance belongs to the trim, and experienced factories tune each from their own consumption records.

Glossary

TERM	MEANING
Consumption ratio	Thread length consumed per unit length of seam for a given stitch type.
Gross	Button trading unit of 144 pieces.
Bill of materials (BOM)	The per-garment list of all fabric, trims and accessories with their quantities.
Trims	Functional non-fabric components sewn in: thread, buttons, zips, labels, interlining.
Accessories	Items added at packing rather than sewing: polybags, tissue, cartons, hang tags.

Key points and summary

- Every trim = per-garment quantity × order quantity × (1 + its own wastage allowance), converted into the supplier's trading unit.
- Thread is calculated by seam length × stitch ratio: the course shirt needs 13.7 m, 15.07 m with wastage; the order needs 37 cones of 5,000 m.
- Buttons: 123,600 pieces = 859 gross; labels 12,240 of each type; polybags 12,120; cartons 500.
- Different trims earn different allowances — thread 10%, buttons 3%, labels 2%, bags 1% — from their different loss mechanisms.

Practice

Answer in your own words; brief model answers follow.

1. A garment has 180 cm of lockstitch and 90 cm of 401 chainstitch seams. Calculate thread per garment before and after a 10% allowance.
2. An order of 20,000 garments needs 4 buttons per garment with 3% wastage. Calculate pieces and gross to buy.
3. Why does the 504 overlock consume roughly five times as much thread per centimetre as the 301 lockstitch?
4. Polybags are bought at 1% allowance but thread at 10%. Give two loss mechanisms that justify the difference.

Answer key

1. 180 × 2.5 + 90 × 4.0 = 450 + 360 = 810 cm = 8.1 m; with allowance 8.1 × 1.10 = 8.91 m.

2. 4 × 20,000 × 1.03 = 82,400 pieces; ÷ 144 = 572.2 → buy 573 gross.
3. The lockstitch locks two threads inside the plies along the seam line. The overlock wraps three threads around the fabric edge in loops on both faces, so each centimetre of seam contains many centimetres of thread.
4. Thread is lost to cone ends, unpicking and re-sewing, thread breaks and shade-lot changeovers; polybags are lost only to occasional tearing or miscounts, and are not consumed by the process itself.
